

```
In [1]: import pandas as pd
df=pd.read_csv("diabetes.csv")
print(df)
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI \
0	6	148	72	35	0	33.6
1	1	85	66	29	0	26.6
2	8	183	64	0	0	23.3
3	1	89	66	23	94	28.1
4	0	137	40	35	168	43.1
..	...	...	...	...	...	...
763	10	101	76	48	180	32.9
764	2	122	70	27	0	36.8
765	5	121	72	23	112	26.2
766	1	126	60	0	0	30.1
767	1	93	70	31	0	30.4

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1
..	...	...	...
763	0.171	63	0
764	0.340	27	0
765	0.245	30	0
766	0.349	47	1
767	0.315	23	0

[768 rows x 9 columns]

```
In [2]: print(df.head())
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI \
0	6	148	72	35	0	33.6
1	1	85	66	29	0	26.6
2	8	183	64	0	0	23.3
3	1	89	66	23	94	28.1
4	0	137	40	35	168	43.1

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

```
In [3]: print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Pregnancies                          768 non-null    int64
1   Glucose                             768 non-null    int64
2   BloodPressure                       768 non-null    int64
3   SkinThickness                      768 non-null    int64
4   Insulin                            768 non-null    int64
5   BMI                                768 non-null    float64
6   DiabetesPedigreeFunction            768 non-null    float64
7   Age                                768 non-null    int64
8   Outcome                             768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
None
```

```
In [4]: print(df.describe())
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin \
count	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458	79.799479
std	3.369578	31.972618	19.355807	15.952218	115.244002
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000	0.000000
50%	3.000000	117.000000	72.000000	23.000000	30.500000
75%	6.000000	140.250000	80.000000	32.000000	127.250000
max	17.000000	199.000000	122.000000	99.000000	846.000000

	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000
mean	31.992578	0.471876	33.240885	0.348958
std	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.078000	21.000000	0.000000
25%	27.300000	0.243750	24.000000	0.000000
50%	32.000000	0.372500	29.000000	0.000000
75%	36.600000	0.626250	41.000000	1.000000
max	67.100000	2.420000	81.000000	1.000000

```
In [9]: cf = pd.read_excel("diabetesex.xlsx")
print(cf)
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	
..	...	...	...	...	...	...	
763	10	101	76	48	180	32.9	
764	2	122	70	27	0	36.8	
765	5	121	72	23	112	26.2	
766	1	126	60	0	0	30.1	
767	1	93	70	31	0	30.4	

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1
..	...	...	...
763	0.171	63	0
764	0.340	27	0
765	0.245	30	0
766	0.349	47	1
767	0.315	23	0

[768 rows x 9 columns]

```
In [10]: import sqlite3
import pandas as pd

df = pd.read_csv("diabetes.csv")

conn = sqlite3.connect("diabetes.db")

df.to_sql("diabetes", conn, if_exists="replace", index=False)
```

Out[10]: 768

```
In [11]: import sqlite3
import pandas as pd
conn = sqlite3.connect("diabetes.db")

query = "SELECT * FROM diabetes"

df = pd.read_sql(query, conn)

print(df)

conn.close()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	
..	...	...	...	...	...	...	
763	10	101	76	48	180	32.9	
764	2	122	70	27	0	36.8	
765	5	121	72	23	112	26.2	
766	1	126	60	0	0	30.1	
767	1	93	70	31	0	30.4	

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1
..	...	...	...
763	0.171	63	0
764	0.340	27	0
765	0.245	30	0
766	0.349	47	1
767	0.315	23	0

[768 rows x 9 columns]

In [13]: `import pandas as pd`

```
url = "diabetes.csv"
```

```
df = pd.read_csv(url)
```

```
print(df.head())
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

In [14]:

```
df.to_csv("diabetes_export.csv", index=False)
```

```
df.to_excel("diabetes_export.xlsx", index=False)
```

```
df.to_json("diabetes_export.json", orient="records")
```

```
import sqlite3
```

```
conn = sqlite3.connect("diabetes.db")
```

```
df.to_sql("diabetes", conn, if_exists="replace", index=False)
```

```
conn.close()
```

In [15]: `import pandas as pd`

```
df = pd.read_csv("diabetes.csv")
```

```
df.to_csv("diabetes_output.csv", index=False)
```

```
print("CSV File Exported")
```

CSV File Exported

In [16]: `import pandas as pd`

```
df = pd.read_csv("diabetes.csv")
```

```
df.to_excel("diabetes_output.xlsx", index=False)
```

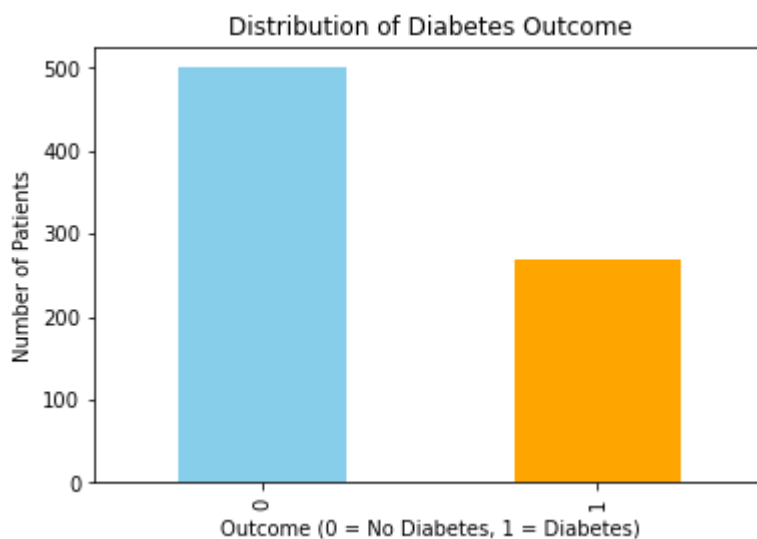
```
print("Excel File Exported")
```

Excel File Exported

```
In [17]: import pandas as pd
df = pd.read_csv("diabetes.csv")
df.to_json("diabetes_output.json", orient="records")
print("JSON File Exported")
```

JSON File Exported

```
In [18]: import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("diabetes.csv")
df["Outcome"].value_counts().plot(kind="bar", color=["skyblue", "orange"])
plt.title("Distribution of Diabetes Outcome")
plt.xlabel("Outcome (0 = No Diabetes, 1 = Diabetes)")
plt.ylabel("Number of Patients")
plt.show()
```



In []: